

Raj Narayan Singh

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Summary

Final-year Computer Science student specializing in AI Engineering and scalable backend development. Currently building production-grade AI systems at Hamath OPC integrating Large Language Models (LLMs), Agentic AI, Computer Vision, and cloud-native backend architectures using Spring Boot, FastAPI, GCP, Docker, LangGraph, and RAG pipelines. Experienced in delivering end-to-end AI products from system design to cloud deployment.

Experience

AI Engineering Intern

Hamath OPC Pvt. Ltd.

May 2026 – July 2026

Remote

- Led the design, development, and deployment of two production-oriented AI products — **EventLens AI** and **RevenueGuard AI** — driving system architecture, backend engineering, AI integration, and cloud deployment from concept to MVP delivery.
- Designed modular backend services and secure RESTful APIs integrating **Computer Vision**, **Agentic AI**, and **Large Language Models (LLMs)** into production business workflows with emphasis on scalability, maintainability, and clean software architecture.
- Implemented cloud-native infrastructure using **Google Cloud Platform (GCP)**, **Google Cloud Storage (GCS)**, Docker, PostgreSQL, and Redis, replacing filesystem-based media management with a provider-agnostic storage architecture supporting secure signed URL generation.
- Collaborated closely with stakeholders throughout the software development lifecycle, translating business requirements into deployable AI applications through iterative design, implementation, testing, and continuous product enhancement.

Technical Skills

Languages: Python, Java, SQL, JavaScript, TypeScript

AI / Machine Learning: Large Language Models (LLMs), Agentic AI, LangGraph, LangChain, Retrieval-Augmented Generation (RAG), FAISS, Sentence Transformers, Computer Vision, Face Recognition, Prompt Engineering

Frameworks: Spring Boot, FastAPI, Next.js, React, Express.js

Cloud & DevOps: Google Cloud Platform (GCP), Google Cloud Storage (GCS), Docker

Databases: PostgreSQL, Redis

Developer Tools: Git, GitHub, Postman, Groq API

Selected Projects

EventLens AI – AI-Powered Event Photography Platform

[GitHub Repository](#)

Spring Boot | Next.js | PostgreSQL | Redis | Google Cloud Storage | Docker | CompreFace

- Evolved **EventLens AI** from a static webpage into a production-ready full-stack AI photography platform — architecting a layered Spring Boot backend separating authentication (JWT + Redis), event management, AI processing, and storage across a 7-container Docker-orchestrated architecture.
- Engineered a dual-stage image processing pipeline — re-encoding all uploads to JPEG 0.95 via canvas and ImageIO to strip HEIC/Snapchat metadata, then routing through **CompreFace** for face detection and recognition at an 80% confidence threshold with automated subject mapping and gallery generation.
- Replaced filesystem-based media storage with a provider-agnostic **StorageService** abstraction over Google Cloud Storage, generating signed URLs dynamically per request with configurable expiry — zero hardcoded URLs persisted to the database.
- Implemented a hybrid fallback architecture where the Next.js frontend degrades gracefully to direct **CompreFace** API calls with localStorage persistence when the Spring Boot backend is unreachable — ensuring zero downtime for end users.

RevenueGuard AI – Enterprise Agentic AI Payment Recovery Platform

[GitHub Repository](#)

Node.js | Express.js | PostgreSQL | Prisma ORM | Groq API | LangGraph | LLaMA 3.3-70B

- Architected an 8-service modular backend for enterprise payment recovery automation — separating Risk Engine, Priority Engine, Commitment Engine, Recommendation Engine, Strategy Engine, and Workflow Engine into independently maintainable service layers over PostgreSQL and Prisma ORM.
- Built a deterministic rule engine combining delayed-days analysis, escalation status tracking, and broken commitment detection to automatically classify 500+ invoices by recovery priority without manual intervention.
- Designed a hybrid AI reasoning pipeline pairing the rule-based recommendation engine with **Groq LLM (LLaMA 3.3-70B)** for personalized risk summaries and recovery strategies, with a **LangGraph**-based multi-agent architecture planned for autonomous end-to-end workflow orchestration.
- Persisted all AI decisions to a structured **AIRecommendation** PostgreSQL table — enabling full auditability, decision traceability, and a queryable AI memory layer supporting future autonomous agent orchestration.

Education

Kalinga Institute of Industrial Technology (KIIT)

Bachelor of Technology in Computer Science & Engineering

Bhubaneswar, Odisha

CGPA: 8.23 / 10 | 09/2022 – 05/2026